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1 Chapter 1

1.1 Edge Connectivity

Lemma 1.1. For any graph G , $\kappa'(G) \leq \delta(G)$, where $\kappa'(G)$ is the vertex connectivity of G : the minimum number of vertices needed to remove to disconnect the graph. $\delta(G)$ is the minimum degree of G . It is the smallest degree among all vertices in G .

Proof. For $\delta(G) = 0$, $\kappa'(G) = 0 \leq \delta(G)$. Assume $\delta(G) \geq 1$. Let v be a vertex of $\delta(G)$. Let F be the set of all edges incident with v . Then $G - F$ is disconnected since v by itself is a component, and there is at least one other vertex in G ($\delta(G) \geq 1$) So $\kappa'(G) \leq |F| = \delta(G)$ $\square$

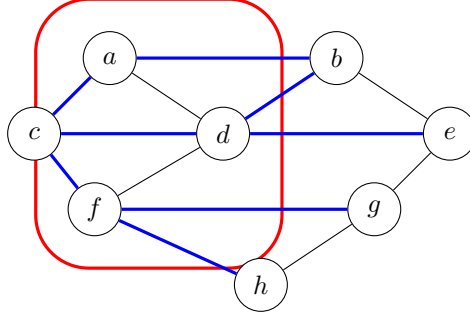
1.2 Cuts and bonds

If $\kappa'(G) = k \geq 1$, then there exists some set of k edges F for which $G - F$ is disconnected. (This is a minimum set)

Definition 1.2 (Cut). For a set $S \subset V(G)$, the **cut induced by S** (or just a cut) is the sets of all edges in G with exactly one end in s . Notation: $\delta_G(S)$ or $\delta(S)$. It is a trivial cut if $S = \emptyset$ or $s = V(G)$

Lemma 1.3. Graph G is connected if and only if every nontrivial cut is non-empty.

Example 1.4.



Let

$$S = \{a, d, f\}.$$

Then

$$\delta(S) = \{ab, ac, db, dc, de, fc, fg, fh\}.$$

In general,

$$\delta(S) = \delta(\bar{S}).$$

where $\bar{S}$ is the complement.

Lemma 1.5. If $F \subset E(G)$ where $G - F$ is disconnected, then F contains a nontrivial cut. If G is not connected, then there is an empty nontrivial cut, F contains it.

Proof. Let H be a component of $G - F$. There is at least one other component. So $\delta_G(V(H))$ is a nontrivial cut in G . It remains to show that $\delta_G(V(H)) \subset F$. Since H is a component of $G - F$, $\delta_{G-F}(V(H)) = \emptyset$. So all edges in $\delta_G(V(H))$ are in F . So F contains a nontrivial cut. Note that when searching for a minimal set of edges that disconnect a graph, we only need to consider nontrivial cuts. $\square$

Definition 1.6 (Bond). A bond is a minimal non-empty cut. Here minimal means no proper subset is a non-empty cut.

Proposition 1.7. Let $F = \delta(S)$ be a cut in a connected graph G . Then F is a bond if and only if $G - F$ has exactly 2 components. In general, for any G , F is a bond if and only if $G - F$ has one more component than G .

Proof. ($\implies$) Suppose $G - F$ has at least 3 components. Say 2 of them are in $G[S]$. Let H be one of them. Then $\delta_G(V(H)) \not\subset F$, contradicting minimality of F .

($\impliedby$) Suppose F is not a bond, so it contains a non-empty cut $\delta(T) \not\subset F$. Then there is an edge e in $F - \delta(T)$ that joins the two components. In $G - \delta(T)$, there is a path from T to e , using an edge that is in $\delta(T)$ but not F . $\square$

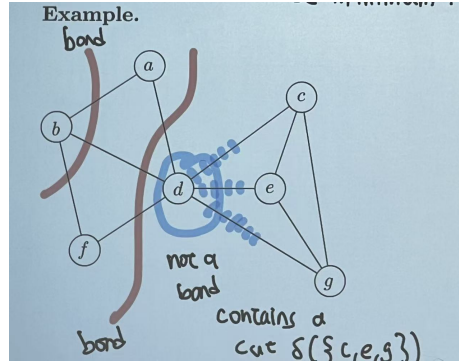


Figure 1: Example of bonds

Proposition 1.8. Every cut is a disjoint union of bonds.

Proof. We prove by induction on the number of edges a cut F . If $|F| = 0$, then F is a disjoint union of 0 bonds. (Inductive hypothesis: For any cut with fewer edges than F , it is a disjoint union of bonds.) Assume F is non-empty. If F is a bond, then we are done. Otherwise, F properly contains a non-empty cut F' by definition. Let $F'' = F - F'$. Then $F'' = F \triangle F'$. By lemma 1.10, F'' is a cut. Also, $|F'| < |F|$ and $|F''| < |F|$ as F' is a non-empty proper subset of F . By induction, F', F'' are both disjoint unions of bonds. Since $F = F' \cup F''$ is a disjoint union, F is also a disjoint union of bonds. $\square$

Definition 1.9 (Symmetric Difference). The **symmetric difference** of two sets S, T is $S \triangle T = (S \cup T) - (S \cap T)$ (ones in exactly 1 of 2 sets)

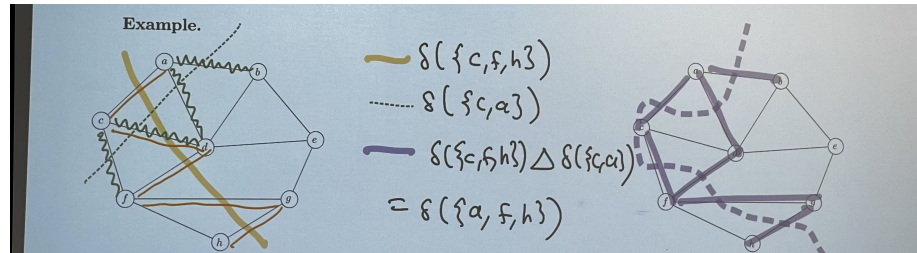


Figure 2: Symmetric Difference

Lemma 1.10. $\delta(S) \triangle \delta(T) = \delta(S \triangle T)$

1.3 Vertex Connectivity

Definition 1.11 (Cut-vertex). A vertex V is a cut-vertex of G if $G - V$ has more components than G . A subset F of vertices is a separating set if $G - F$ is disconnected. (For a disconnected graph, $\emptyset$ is a separating set.)

Example 1.12. The connectivity for the below 2 graphs are $\kappa(G_1) = 1, \kappa(G_2) = 3$

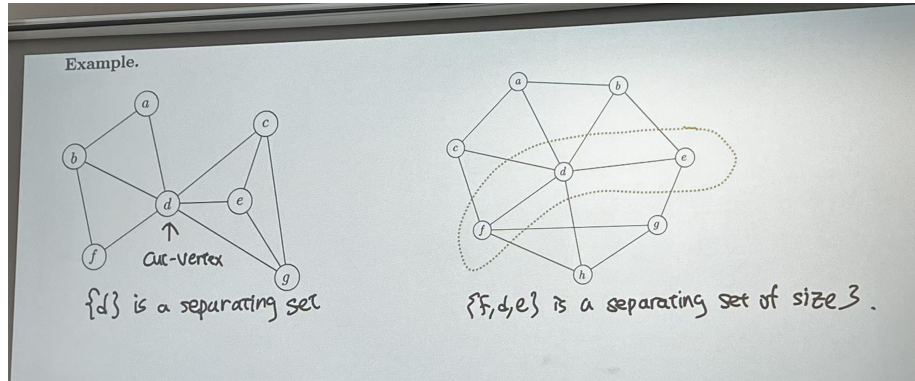


Figure 3: Example of separating set

Definition 1.13 (K-connected). A graph is k -connected if it has at least $k + 1$ vertices and does not have a separating set of size at most $k - 1$. (i.e. Remove $k-1$ vertices, the graph is still connected) The connectivity of G is the largest k for which G is k -connected. Notation: $\kappa(G)$

Note

- 1-connected $\iff$ connected with ≥ 2 vertices.
- 2-connected $\iff$ connected with no cut-vertex with ≥ 3 vertices. (remove 2 vertices it's still connected)
- K_n cannot be disconnected. Using the vertex size requirement, $\kappa(K_n) = n - 1$

Lemma 1.14 (Expansion Lemma). Let G be a k -connected graph. Let G' be obtained from G by adding a vertex V and at least k edges joining V with distinct vertices of G . Then G' is k -connected.

Proof. Let X be any subset of at most $k-1$ vertices in G' . We break into 2 cases:

1. Suppose $v \in X$. Then $G' - X$ is the same graph as removing at most $k-2$ vertices from G . [$G' - X = G - (X - v)$] Since G is k -connected, $G' - X$ is connected. So X is not a separating set.
2. Suppose $v \notin X$. Since $|X| \leq k - 1$ and G is k -connected, $G - X$ is connected. Since v has k neighbours in G , at least one of them, say w is not in X . Then for any $y \in V(G) - X$, take the path from y to w in $G - X$, follow by edge wv . So a vy -path exists for all $y \in V(G) - X$. So $G' - X$ is connected, X is not a separating set.

□

1.4 *Note

$$\kappa(G) \leq \kappa'(G)$$

Theorem 1.15 (Whitney's Theorem). For any graph G , $\kappa(G) \leq \kappa'(G) \leq \delta(G)$

Proof. By lemma 1.1, $\kappa'(G) \leq \delta(G)$. we now prove $\kappa(G) \leq \kappa'(G)$. For a complete graph K_n , $\kappa(G) = \kappa'(G) = n - 1$. For a disconnected graph G , $\kappa(G) = \kappa'(G) = 0$. We assume G is connected but not complete. Let F be a minimum nontrivial cut in G . So $|F| = \kappa'(G)$. Then F is a bond. By prop, $G-F$ has exactly 2 components, say H_1, H_2 . We have 2 cases:

1. Suppose there exists a vertex V not incident to any edges in F . WLOG, say v is in H_1 . Let S be the set of all vertices in H_1 , incident with some edge in F . Then $|S| \leq |F|$. Also, S is a separating set as it separates v from $V(H_2)$ (check details). So $\kappa(G) \leq |S| \leq |F| = \kappa'(G)$.
2. Suppose all vertices in G are incident to some edge in F . Since G is not complete, there exist $u, v \in V(G)$ that are not adjacent. WLOG, say $u \in V(H_1)$. Let S be all neighbours of u . We see that $v \notin S$. Then S is a separating set, as it separates u from v . We know count the size of S . Let $y \in S$. If $y \in V(H_2)$, then uy is an edge in F . If $y \in V(H_1)$, then by assumption, y is incident to some edge in F . All such edges are distinct in F . So $|S| \leq |F|$.

□

Side quest: Induction

We want to prove a statement depending on an integer parameter k .

- Base case.
- Assume the statement is true for everything up to $k-1$.
- Prove that the statement is true for k .

Note that it is better to start with a large vertices then remove rather than start with a small vertices then add since there will be cases missed (i.e. may not include all cases).

Framework for induction in graph theory

Goal: Prove that "if some conditions hold, then some results hold", with induction on a parameter n .

- Base case: Do at least one, do more if the induction step does not apply to them.
- Hypothesis: Assume that the results hold for any graph satisfying the conditions with parameter $\leq n - 1$

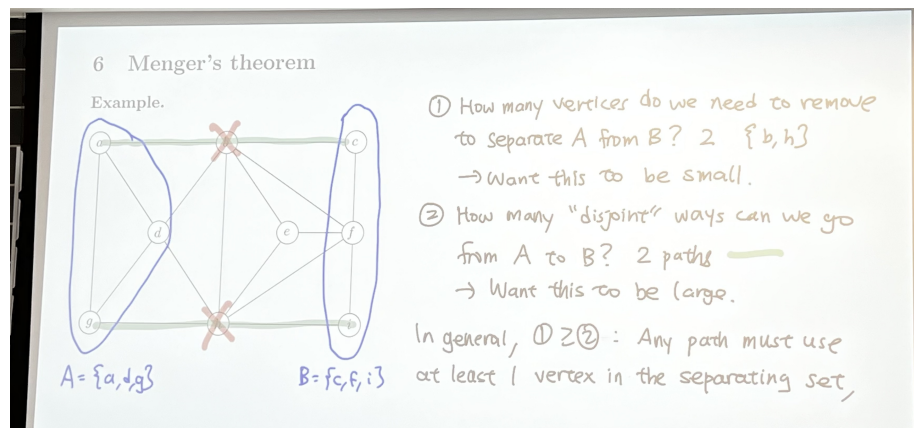


Figure 4: Example of Menger's Theorem

- Induction step: Start with an arbitrary graph satisfying the conditions with parameter n
- Modify to get a new graph H with parameter $< n$ (usually by removing or contracting something, or taking a proper subgraph). Check that H satisfies the conditions, and by induction the results hold for H
- Use the results on H to derive that the results hold for G .

1.5 Menger's Theorem

So the number of disjoint paths is $\leq$ size of separating set.

Definition 1.16 (A,B-path). An **A,B-path** is a path with one end in A , the other end in B , and at most one vertex in A (or B) is in the path. (If A and B intersect, then a single vertex is the A-B path)

Definition 1.17 (Disjoint A,B-paths). A set of A, B paths is disjoint if no vertex is in two or more paths.

Definition 1.18 (A,B-separator). Let $A, B \subset V(G)$ (not necessarily disjoint). A set of vertices X is an A,B-separator if $G-X$ does not have any A,B-paths. (A and B are A,B-separators)

Theorem 1.19 (Menger's Theorem). Let G be a graph, and $A, B \subset V(G)$. Then the minimum size of an A,B-separator is equal to the maximum size of a set of disjoint A,B-paths

Proof. (Easy direction): We need to prove the minimum separator $\geq$ maximum paths. If there are k disjoint A,B-paths, then any A,B-separator must include at least one vertex from each path. (otherwise an A,B-path remains after removal, A,B not separated.) $\square$

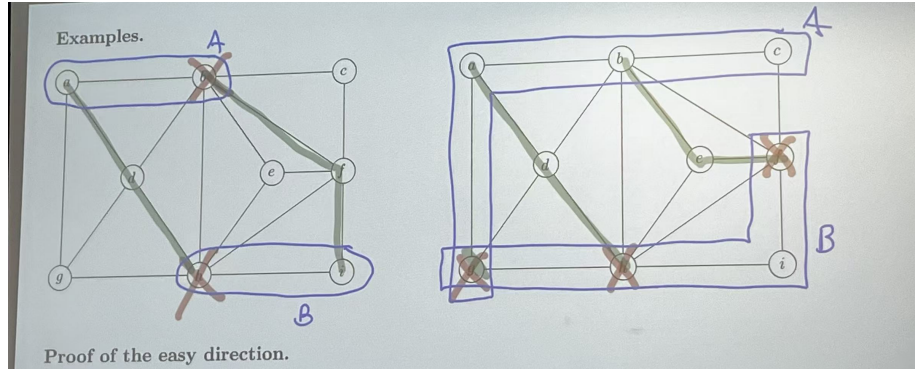


Figure 5: Example of Menger's Theorem

Goal for the hard direction:

Let k be the size of a minimum A, B -separator. Need to show that there exists a set of k disjoint A, B -paths.

An easy but wrong proof of the hard direction

Induction on $|E(G)|$. Let X be a min A, B -separator. Let Y be an A, X -separator. Then k is also an A, B -separator. This means size of minimum A, X -separator is k . By induction, there are k disjoint A, X -paths. Same argument, k disjoint X, B -paths. Thus, there are k disjoint A, B -paths.

Problem: If $X = B$, then the first part is the whole graph. Hence we cannot apply induction.

Useful fact

Let S be an A, B -separator. Then every A, B -path must use at least one vertex in S .

Proof. Prove by induction on $|E(G)|$. When $|E(G)| = 0$, the minimum A, B -separator has size $|A \cap B|$. Take the trivial paths for vertices in $A \cap B$ to find $|A \cap B|$ disjoint A, B -paths.

Assume $|E(G)| \geq 1$. Suppose the size of a minimum A, B -separator in G is k . Let $e = xy \in E(G)$. Consider $G - e$. If the size of a min A, B -separator in $G - e$ is k , then by induction, there exists a set of k disjoint A, B -paths in $G - e$, which are also in G . We are done.

Assume there is a min A, B -separator S of $G - e$ of size at most $k - 1$. We claim that $|S| = k - 1$. Now S is not an A, B -separator in G (too small, $< k$). So every A, B -path in $G - S$ must use e . Note that every A, B -paths in $G - S$ must go through x, y in the same order, say x first and then y . Then $S \cup \{x\}, S \cup \{y\}$ are A, B -separators in G (they remove e). So $|S \cup \{x\}| \geq k$. But $|S| \leq k - 1$. So $|S| = k - 1$. $\square$

Proof of the hard direction continued

Let

$$S = \{v_1, \dots, v_{k-1}\}, \quad S_x = S \cup \{x\}, \quad S_y = S \cup \{y\}.$$

Recall that both S_x and S_y are A, B -separators in G .

Let T be a minimum A, S_x -separator in $G - e$. Then T is also an A, B -separator in G . Hence

$$|T| \geq k.$$

But

$$|S_x| = k,$$

so any A, S_x -separator has size at most k . Therefore

$$|T| = k.$$

By induction, there exist k disjoint A, S_x -paths in $G - e$. Since $|S_x| = k$, there is one path ending at each vertex of S_x . Note that none of these paths include y .

Let

$$P_1, \dots, P_{k-1}, P_x$$

be the disjoint A, S_x -paths ending in

$$v_1, \dots, v_{k-1}, x$$

respectively.

Using a similar argument, we can obtain k disjoint S_y, B -paths

$$Q_1, \dots, Q_{k-1}, Q_y$$

starting at

$$v_1, \dots, v_{k-1}, y$$

respectively.

Since the P_i 's and Q_i 's are disjoint except at $v_1, \dots, v_{k-1}$, we can get k disjoint A, B -paths:

$$P_1 + Q_1, \dots, P_{k-1} + Q_{k-1}, P_x + xy + Q_y.$$

This completes the proof.

2 Other Versions of Menger's Theorem

Definition 2.1 (*x, y -separator*). Let $x, y \in V(G)$ be distinct and non-adjacent. An x, y -separator is a set of vertices S such that x and y are in different components of $G - S$.

Definition 2.2 (*Internally-disjoint x, y -paths*). A set of x, y -paths are *internally-disjoint* if the only vertices they have in common are x and y .

Theorem 2.3 (Menger's Theorem, vertex version). Let x, y be distinct non-adjacent vertices in G . Then the size of a minimum x, y -separator is equal to the maximum size of a set of internally-disjoint x, y -paths.

Proof sketch. Let A and B be the sets of all neighbours of x and y in G , respectively. Apply the set version of Menger's Theorem.

Minimum A, B -separators correspond to minimum x, y -separators, and disjoint A, B -paths can be extended to internally-disjoint x, y -paths. $\square$

Theorem 2.4 (Menger's Theorem, edge version). Let x, y be distinct vertices of G . Then the minimum size of a cut that separates x and y is equal to the maximum size of a set of edge-disjoint x, y -paths.

Definition 2.5 (Line graph). The line graph of G , denoted $L(G)$, is the graph with vertex set

$$V(L(G)) = \{v_e : e \in E(G)\},$$

where $v_e v_f$ is an edge if and only if e and f share a vertex in G .

Sketch proof of Theorem 12. Form the line graph $L(G)$. Let A and B be vertices in $L(G)$ corresponding to all edges incident to x and y , respectively. Apply the set version of Menger's Theorem.

Then

$$A, B\text{-paths in } L(G) \iff \text{edge-disjoint } x, y\text{-paths in } G,$$

and

$$A, B\text{-separators in } L(G) \iff \text{cuts that separate } x, y \text{ in } G.$$

$\square$

3 Corollaries of Menger's Theorem

3.1 Characterization of k -connected and k -edge-connected graphs using paths

Corollary 3.1 (Characterization of k -connected graphs). A graph G is k -connected if and only if there exist k internally-disjoint x, y -paths for any distinct $x, y \in V(G)$.

Proof. ($\Rightarrow$) Assume G is k -connected. Let $x, y \in V(G)$ be distinct.

If x and y are not adjacent, then any minimum x, y -separator has size at least k , since G is k -connected. By Menger's Theorem, vertex version, there exist k internally-disjoint x, y -paths.

If x and y are adjacent, then $G - xy$ is $(k - 1)$ -connected. Hence any x, y -separator in $G - xy$ has size at least $k - 1$. By Menger's Theorem, there exist $k - 1$ internally-disjoint x, y -paths in $G - xy$. Together with the edge xy , we have k internally-disjoint x, y -paths in G .

($\Leftarrow$) Assume that for any distinct $x, y \in V(G)$, there exist k internally-disjoint x, y -paths. This assumption implies that

$$|V(G)| \geq k + 1.$$

Let $S \subseteq V(G)$ with

$$|S| \leq k - 1.$$

Take any $x, y \in V(G) - S$ with $x \neq y$. By assumption, there are k internally-disjoint x, y -paths in G . At least one of these paths remains in $G - S$. Therefore $G - S$ is connected. Hence G is k -connected. $\square$

Corollary 3.2. If G is k -connected, then for any $A, B \subseteq V(G)$ with

$$|A|, |B| \geq k,$$

there exist k disjoint A, B -paths.

The converse is not true.

Corollary 3.3 (Characterization of k -edge-connected graphs). A graph G is k -edge-connected if and only if there exist k edge-disjoint x, y -paths for any distinct $x, y \in V(G)$.

3.2 Corollaries of Menger's theorem

Definition 3.4 (k-fan). Let $v \in V(G)$, and $X \subset V(G) - \{v\}$. A k -fan from v to X is a set of k internally-disjoint paths that start with v , end at distinct vertices in X , and no internal vertices are in X .

Theorem 3.5 (Fan lemma). Let G be k -connected, $v \in V(G)$, $X \subset V(G) - \{v\}$, where $|X| \geq k$. Then there exists a k -fan from v to X .

Proof. Let Y be the set of all neighbours of v . Then $|Y| \geq k$ since $\delta(G) \geq k$ by Whitney's. By corollary 3.2, there exist k disjoint Y, X -paths. Such paths do not include v , otherwise they contain at least 2 vertices of Y . $\square$

Corollary 3.6. Let G be a k -connected graph with $k \geq 2$. Let $X \subset V(G)$ where $|X| = k$. Then G has a cycle containing all vertices in X .

Proof. We prove by induction on k . When $k = 2$, G is 2-connected. For any $x, y \in V(G)$, by corollary 3.1, there exist 2 i.d. x, y -paths. Together they form a cycle containing x, y . Assume $k \geq 3$. Suppose G is k -connected, and $X \subset V(G)$ where $|X| = k$. [Hypothesis: $(k-1)$ -connected, $|Y| = k-1$ implies there exist a cycle containing Y .] Since G is k -connected, G is $(k-1)$ -connected.. Let $v \in X$. By induction, there is a cycle C in G that containing $X - \{v\}$. If C contains v , then C contains X , and we are done. Assume v is not in C . We label the $k-1$ vertices of $X - \{v\}$ on C in cyclic order, $v_1, \dots, v_{k-1}$. They partition C into $k-1$ i.d. paths, we call them segments (v_1 to v_2 , v_2 to v_3 , etc.)

Suppose C has at least k vertices. By Fan Lemma, there is a k -fan from v to $V(C)$. Since there are k paths in the fan and $k-1$ segments on C , at least 2

paths P_1, P_2 end in the same segment, say they end at vertices w_1, w_2 . Let P_3 be the w_1, w_2 -paths in C in the segment. Then P_3 does not have any vertices of X as internal vertices. Then $C - P_3 + P_1 + P_2$ is a cycle in G containing X .

Suppose C has $k - 1$ vertices. Then there is a $(k - 1)$ -fan from v to $V(C)$, ending in all vertices of C . Remove an edge xy in C , add the 2 paths in the fan ending in x, y . This gives a cycle containing X . $\square$

4 Structure of 2-connected graphs

4.1 Characterization via cycles

Theorem 4.1 (Cycle characterization of 2-connected graphs). Let G be a graph with at least 3 vertices. Then the following are equivalent:

1. G is 2-connected
2. For any distinct $u, v \in V(G)$, there is a cycle containing u, v .
3. $\delta(G) \geq 1$, and for any distinct edges $e, f \in E(G)$, there is a cycle containing e, f .

Proof. $2 \implies 1 \implies 3 \implies 2$

$2 \implies 1$: A cycle containing u, v consists of 2 i.d. u, v -paths. By corollary 3.1, G is 2-connected.

$1 \implies 3$: Since G is 2-connected, $\delta(G) \geq 2$ by Whitney. Let $e = uv, f = xy$ be 2 distinct edges in G . Let $A = \{u, v\}, B = \{x, y\}$. Since G is 2-connected and $|A|, |B| \geq 2$, by corollary 3.2, there exist 2 disjoint A, B -paths P_1, P_2 . Then $P_1 + e + P_2 + f$ is a cycle in G containing e, f .

$3 \implies 2$ Let $u, v \in V(G)$ be distinct vertices. If u, v are not adjacent, then there exist distinct edges e, f incident to u, v respectively. ($\delta(G) \geq 1$). By assumption, there exists a cycle containing e, f which contains u, v . If uv is an edge, then there exist another vertex w (since $|V(G)| \geq 3$). And w is incident with some edge h (since $\delta(G) \geq 1$) distinct from uv . By assumption, there is a cycle containing h and uv , which contains u, v .

Note that the last direction will fail if $|V(G)| < 3$ vertices or if $\delta(G) < 1$. $\square$

Definition 4.2 (Ear). Let F be a subgraph of G . An *ear* of F is a nontrivial path in G where only the two endpoints of the path are in F , and no edge of the path is in F .

Definition 4.3 (Ear decomposition). An ear decomposition of G is a sequence of graphs $(G_0, G_1, \dots, G_k)$ where

- G_0 is a cycle in G
- $G_k = G$ and
- For $0 \leq l \leq k - 1$, $G_{l+1} = G_l + P_l$ where P_l is an ear of G_l .

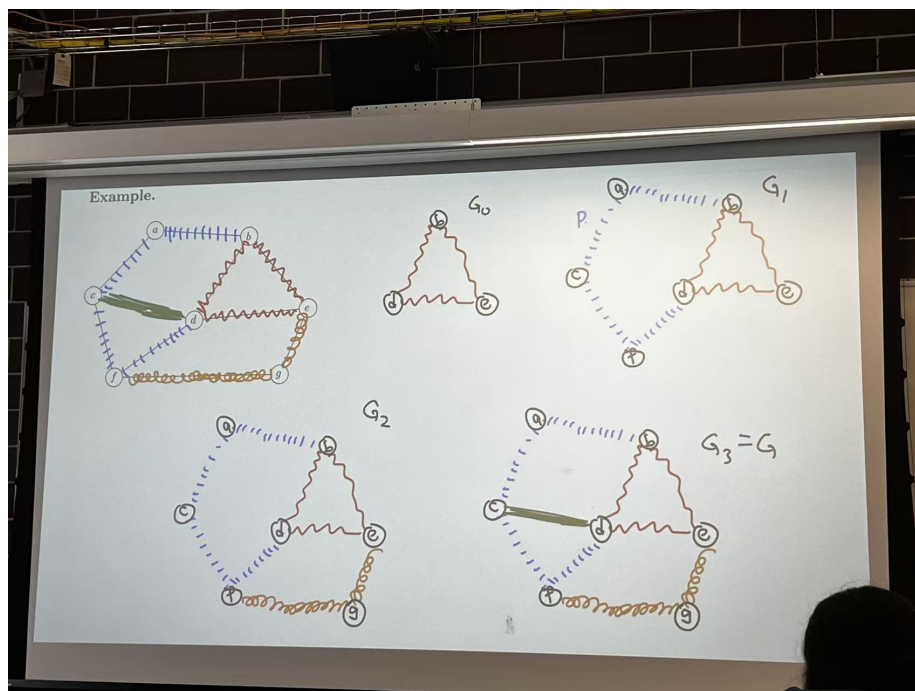


Figure 6: Example of ear decomposition

Theorem 4.4 (Ear decomposition characterization of 2-connected graphs). A graph G is 2-connected if and only if G has an ear decomposition.

Proof. ($\implies$) Let G be 2-connected, let c be any cycle in G . Let $G_0 = c$. For $i \geq 0$, we inductively construct G_{i+1} from G_i as follows:

If $G_i = G$, then we are done. Otherwise there exists an edge $e = uv$ in $E(G) - E(G_i)$. Since G is 2-connected, by Corollary, there exist 2 disjoint paths between $\{u, v\}$ and $V(G_i)$. Let Q_1, Q_2 be these paths. Then $P_i = Q_1 + e + Q_2$ is an ear of G_i (only the two endpoints are in G_i). Set $G_{i+1} = G_i + P_i$. At termination, say after k steps, $(G_0, \dots, G_k)$ is an ear decomposition of G .

($\impliedby$) Suppose $(G_0, \dots, G_k)$ is an ear decomposition of G , where $G_{i+1} = G_i + P_i$, where P_i is an ear of G_i . We inductively show that G_i is 2-connected. Since G_0 is a cycle, G_0 is 2-connected. Assume G_i is 2-connected. (Show G_{i+1} is 2-connected). Let u, v be the two endpoints of P_i . Let $x, y \in V(G_{i+1})$. We show that x, y are in a cycle in G_{i+1} .

Notes: An ear decomposition can start with any cycle. All intermediate graphs are 2-connected.

Fun fact: For G , the number of ears in any ear decomposition is fixed, $|E(G)| - |V(G)|$ $\square$

Definition 4.5 (Orientation). An orientation of G is obtained by adding a direction to each edge uv , either u to v or v to u , denoted uv or vu .

Definition 4.6 (Directed u,v-path). A directed u, v -path has the form $u, v_1, v_2, \dots, v_k, v$ where $uv_1, \dots, v_kv$ are directed edges.

Definition 4.7 (Strong orientation). A strong orientation is an orientation where there exists a directed u, v -path for any ordered $u, v \in V(G)$

Theorem 4.8. If G is 2-connected, then G has a strong orientation.

5 Blocks

Definition 5.1. A block is a maximally connected subgraph with no cut-vertices.

What does a block look like?

- isolated vertex
- cut-edge
- 2-connected

5.1 Properties of blocks

Proposition 5.2. Every block is an induced subgraph.

Proof. If a subgraph H has no cut-vertices, then adding an edge to H cannot create a cut-vertex. If H is connected, then it is still connected after adding an edge. By maximality, H must include all edges in $G[V(H)]$ $\square$

Proposition 5.3. If c is a cycle, then $V(C)$ is in the same block.

Proof. A cycle is connected with no cut-vertices. By maximality, these vertices must be in the same block. $\square$

Proposition 5.4. Two blocks in a graph share at most one vertex.

Proof. Suppose blocks B_1, B_2 share at least 2 vertices. Let B be the graph with $V(B) = V(B_1) \cup V(B_2)$ and $E(B) = E(B_1) \cup E(B_2)$. [We want to show that B has no cut-vertices.] We can easily show that B is connected. Let $x \in V(B)$ [Show $B - x$ is connected.] Since B_1, B_2 have no cut-vertices, $B_1 - x$ and $B_2 - x$ are connected. [x may or may not be in B_1 or B_2]. Since B_1, B_2 share at least 2 vertices, at least one such vertex y is not x . For any $u, v \in V(B) - \{x\}$, if u, v are both in B_1 or in B_2 , then we are done. Otherwise, WLOG assume $u \in V(B_1)$ and $v \in V(B_2)$. Then we take a u, y -path in $B_1 - x$ followed by a y, v -path in $B_2 - x$ to get a u, v -path in $B - x$. So B has no cut-vertices, and B has at least 1 vertex not in $V(B_1) \cap V(B_2)$. This contradicts the maximality of B_1 or B_2 . $\square$

Proposition 5.5. A vertex in 2 or more blocks is a cut-vertex

Proposition 5.6. If u, v are in the same block B , then all u, v -paths in G are entirely in B . (Otherwise there is a cycle, must be in the same block)

Note

Any edge is in exactly one block (cannot be in 2 blocks by proposition, it must be in some blocks as it does not a cut-vertex) So the blocks partition $E(G)$.

5.2 Block graph

Definition 5.7 (Block graph). The **block graph** of G is a bipartite graph with bipartition (B, C) , where B is the set of all blocks, and C is the set of all cut-vertices, and a block $b \in B$ is adjacent to $v \in C$ if and only if v is in the block b .

Proposition 5.8. The block graph is a forest (No cycles)

Proof. sketch: If there is a cycle, then it gives cycle in G . That is in multiple blocks, contradicting proposition. $\square$

When blocks are useful

G has this property if and only if every block of G has this property, which reduces problems down to blocks (mostly 2-connected)

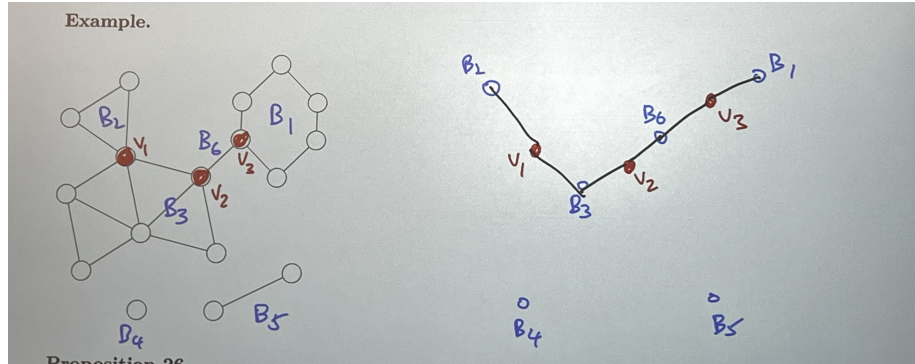


Figure 7: Example of a block graph

5.3 Strong orientation revisited

Recall that a 2-connected graph has a strong orientation. But these are not the only graphs with a strong orientation.

Theorem 5.9. A connected graph G has a strong orientation if and only if every block of G has a strong orientation.

Proof. ($\implies$) Consider a strong orientation of G . Let B be the block of G . Keep the same orientation of G for B . Let $x, y \in V(B)$. There is a directed x, y -path in G . Its underlying undirected path is in B by prop. So this is a directed x, y -path in B , so B has a strong orientation.

($\impliedby$) Suppose every block has a strong orientation. Keep the same orientation for G . Let $x, y \in V(G)$. If x, y are in the same block, then a directed x, y -path exists inside the block. Suppose x, y are in different blocks. Since G is connected, there is a path $B_1, v_1, B_2, v_2, \dots, v_{k-1}, B_k$ in the block graph where $x \in V(B_1), y \in V(B_k)$. Since each B_i has a strong orientation, we can find directed paths from x to v_1 in B_1 , v_1 to v_2 in B_2 , ..., v_{k-1} to y in B_k . This gives a directed x, y -path in G . $\square$

Corollary 5.10. A graph with at least one edge has a strong orientation iff it is 2-edge-connected. (Each block is 2-connected, a cut-edge has no strong orientation.)

6 Structure of 3-connected graphs

We characterize 3-connected graphs based on contradictions.

Theorem 6.1. Let G be a 3-connected graph with at least 5 vertices. Then G has an edge e such that G/e is 3-connected. (Remove multiple edges after contractions)

Corollary 6.2. A graph G is 3-connected if and only if there exists a sequence of graphs $G_0, G_1, \dots, G_k$ where

1. $G_0 = K_4$
2. $G_k = G$
3. for each $0 \leq i \leq k-1$, there exists an edge xy in G_{i+1} such that $d_{G_{i+1}}(x), d_{G_{i+1}}(y) \geq 3$ and $G_i = G_{i+1}/xy$

Proof. sketch: ($\implies$) Apply the previous theorem by induction. Only 3-connected graphs with 4 vertices is K_4 . $\square$

Proof. (proof of theorem) : Let G be 3-connected. Suppose by way of contradiction, for any edge xy , G/xy is not 3-connected. So there exists a separating set S of G/xy where $|S| \leq 2$. Let V_{xy} be the contracted vertex. Then S must contain V_{xy} (otherwise it separates G). Also, $|S| = 2$, for otherwise $\{x, y\}$ is a separating set in G . So $S = \{V_{xy}, z\}$ for some $z \neq x, y$. Let $T = \{x, y, z\}$ in G . So $(G/xy) - S = G - T$. Then T is a minimum separating set in G . Then every vertex in T is adjacent to some vertex in every component of $G - T$.

Among all possible choices of edge xy , vertex z to form $T = \{x, y, z\}$, and a component C of $G - T$, we pick one that minimizes $|V(C)|$. Let v be a neighbour of z in C . Now G/zv is not 3-connected, so there exists w such that $T' = \{z, v, w\}$ is a minimum separating set of G . [Note: w could be x or y .] Since xy is an edge, there exists a component D of $G - T'$ that does not contain x nor y . ($u \neq z$, u is in a component without x, y) For any other vertices in D , there exists a path to u in D that does not use x, y, z . This path still exists in $G - T$, more specifically in C . So $V(D) \not\subseteq V(C)$ since $v \in C$ but $v \in D$. This contradicts the minimality of C . $\square$

7 Edge space

Definition 7.1 (Edge space). The edge space of G is the set of all vectors in $GF(2)$, denoted $\mathcal{E}(G)$. ($GF(2)$ acts like $\mathbb{Z}_2 = \{0, 1\}$). An element of $\mathcal{E}(G)$ is a 0,1-vector indexed by the edges $E(G)$

Note that edge space is 2-base. That is $1 + 1 = 0$

Review of linear algebra

- Subspace: A non-empty set of vectors S in $\mathcal{E}(G)$ is a subspace if it is closed under addition.
- Linear independence: $X \subset \mathcal{E}(G)$ is linearly independent if there is no non-empty subset of X that sum to the zero vector. Dependent otherwise.

Let S be a subspace of $\mathcal{E}(G)$. Let $X \subset S$.

- Span: X spans S if every element of S is the sum of some elements of X .
- Basis: X is a basis of S if X is linearly independent and spans S .
- Dimension: The dimension of S is the size of its basis, $\dim S$.

Let S be a subspace of dimension k . Let $X \subset S$, then X is a basis if 2 of the following holds:

1. X is independent
2. X spans S
3. $|X| = k$

Let $E(G) = \{e_1, \dots, e_m\}$. The set $\{\{e_1\}, \{e_2\}, \dots, \{e_m\}\}$ is a basis of $\mathcal{E}(G)$. So $\dim \mathcal{E}(G) = |E(G)|$

8 Subspaces of the edge space

8.1 Cut space

Definition 8.1 (Cut space). The cut space of G is the set of all cuts in G . Notation : $B(G)$.

Proposition 8.2. $B(G)$ is a subspace of $\mathcal{E}(G)$

Proof. By lemma, the symmetric difference of 2 cuts is a cut. □

Proposition 8.3. The set of all bonds spans $B(G)$.

Proof. By proposition, all cuts are disjoint unions of bonds. □

8.2 Cycle space

Definition 8.4. The cycle space of G is the subspace of $\mathcal{E}(G)$ spanned by the cycles of G , denoted $\mathcal{C}(G)$

Note

Elements of $\mathcal{C}(G)$ might be cycles.

Definition 8.5 (Orthogonal). $F_1, F_2 \in \mathcal{E}(G)$ are **orthogonal** if $|F_1 \cap F_2|$ is even.

Definition 8.6 (Orthogonal complement). If S is a subspace of a vector space $\mathbb{V}$, then the **orthogonal complement** of S is $S^\perp = \{v \in \mathbb{V} : v.w = 0, \forall w \in S\}$

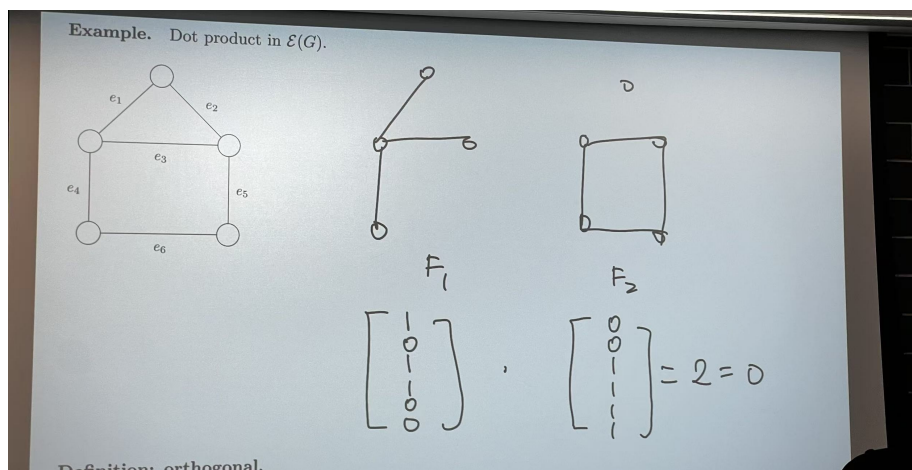


Figure 8: Example of a dot product

Some facts about finite-dimensional orthogonal complements in general

- $S^\perp$ is a subspace of $\mathbb{V}$
- $(S^\perp)^\perp = S$
- $\dim(S) + \dim(S^\perp) = \dim(\mathbb{V})$

check: $\dim B(G) + \dim \mathcal{C}(G) = [|V(G)| - k] + [|E(G)| - |V(G)| + k] = |E(G)| = \dim \mathcal{C}(G)$

Theorem 8.7. $B(G)^\perp = \mathcal{C}(G)$ and $\mathcal{C}(G)^\perp = B(G)$

Proof. It suffices to show that $B(G)^\perp = \mathcal{C}(G)$.

($\subset$) Let $F \in B(G)^\perp$. So F is orthogonal to all cuts of the form $\delta(\{v\})$ for all $v \in V(G)$. Then $|F \cap \delta(\{v\})|$ is even, so F is even. So $F \in \mathcal{C}(G)$ by proposition.

(Other direction) Let $F \in \mathcal{C}(G)$. Let $D \in \delta(X)$ for some $X \subset V(G)$. (Lemma : $\delta(S) + \delta(T) = \delta(S \triangle T)$) By lemma, $F \cap \delta(x) = \sum_{v \in X} (F \cap \delta(v))$. Since $F \in \mathcal{C}(G)$, $|F \cap \delta(v)|$ is even. So $|F \cap \delta(X)|$ is also even. So $F \in B(G)^\perp$. $\square$

Miscellaneous notes on vector spaces in graphs

- If S is a subspace of $\mathcal{C}(G)$ of dimension k , then $|S| = 2^k$ (Elements of S are unique sums of a subsee of the basis)
- It is not true that a basis of a subspcae S of $\mathcal{E}(G)$ and a basis of $S^\perp$ combined spans $\mathcal{E}(G)$. ($S \cap S^\perp$ might have non-zero elements)

- An application; Kirchhoff's circuit laws: Given an electric circuit, the sum of voltages around any cycle must be 0. This is true if and only if the law holds for a basis of the cycle space.

9 Planar graphs

Definition 9.1 (Planar graph, planar embedding). A graph is planar if it can be drawn on the plane so that edges only intersect at common endpoints. The drawing itself is planar embedding.

Definition 9.2 (Plane graph). We use plane graph to refer to a planar graph along with one possible planar embedding. We use points and lines to refer to representations of vertices and edges in the embedding, respectively.

9.1 Some topological details

Definition 9.3 (Curve). A curve is a continuous image of a closed unit line segment. It is **simple** if it does not intersect itself (mapping is injective).

Definition 9.4 (Closed curve). A **closed curve** is a continuous image of a unit circle.

Note

An embedding of an edge is a simple curve. An embedding of an cycle is a simple closed curve.

Definition 9.5 (Arcwise-connected). A set of points is arcwise-connected if any two points in the set are the endpoints of a curve entirely inside the set.

Theorem 9.6 (Jordan Curve Theorem). Any simple closed curve on the plane partitions the rest of the plane into 2 disjoint arcwise-connected open sets, one inside and one outside.

Definition 9.7. Given a simple closed curve C , the open set of points inside C is the interior, denoted $int(C)$. (does not include C). The open set of points outside of C is the exterior, denoted $ext(C)$. If we include C in the sets, then they are closed sets, denoted by $Int(C)$, $Ext(C)$ Note: Any curve that joins a point in $int(C)$ with a point in $ext(C)$ must intersect C .

9.2 Background for Kuratowski's theorem

Recall that K_5 and $K_{3,3}$ are not planar.

Definition 9.8 (Subdivision). A **subdivision** of a graph G is obtained from G by replacing each edge with a new path whose interior vertices are new to G and whose length is at least 1.

Equivalently, we introduce new vertices of degree 2 on some edges.

Example 9.9. A subdivision of K_4 is obtained by replacing some edges of K_4 with paths. The original vertices of K_4 remain, and the new vertices on the paths have degree 2.

Proposition 9.10 (Proposition 45). A graph G is planar if and only if every subdivision of G is planar.

Sketch proof. An edge and a path are both drawn as simple curves. If G has a planar embedding, replacing an edge by a path gives a planar embedding of the subdivision. Conversely, if every subdivision of G is planar, then in particular G itself is planar. $\square$

Corollary 9.11 (Corollary 46). Any graph containing a subdivision of K_5 or $K_{3,3}$ is not planar.

Proof. Since K_5 and $K_{3,3}$ are not planar, any subdivision of them is not planar by Proposition 45. Hence any graph containing such a subdivision cannot be planar. $\square$

Theorem 9.12 (Kuratowski's theorem). A graph is planar if and only if it does not contain a subdivision of K_5 or $K_{3,3}$.

Proof idea. The forward direction follows from Corollary 46. The converse direction is hard and will be proved later. $\square$

10 Faces

10.1 Facial expressions

Definition 10.1 (Face). A **face** of a planar embedding is a maximal subset of points in the plane that is arcwise-connected and does not contain any part of the embedding.

Definition 10.2 (Outer face). Every planar embedding has an unbounded face, called the **outer face**.

Definition 10.3 (Incident). A face is **incident** with the vertices and edges represented by the points and lines on its boundary.

Definition 10.4 (Adjacent faces). Two faces are **adjacent** if they share at least one edge on their boundaries.

Proposition 10.5 (Proposition 47). Let f be a face of a plane graph G . Then there is an embedding of G with the boundary of f as the boundary of the outer face, and all faces have the same boundary vertices and edges as before.

Sketch proof. Use stereographic projection. Draw the plane graph on a sphere with the chosen face on top, then project from the top of the sphere to the plane below. This makes the chosen face become the outer face. $\square$

10.2 Faces and connectivity

Theorem 10.6 (Theorem 48). In a 2-connected plane graph, every face boundary is a cycle.

Proof. By the ear decomposition theorem, there is an ear decomposition

$$(G_0, G_1, \dots, G_k)$$

for G . We treat each G_i as the plane graph whose drawing is part of the plane graph G . We prove inductively that every face boundary of each G_i is a cycle.

For G_0 , it is a cycle, so there are two faces, each bounded by this cycle.

Suppose for some $i \geq 0$ that every face boundary of G_i is a cycle, and suppose

$$G_{i+1} = G_i + P_i,$$

where P_i is an ear of G_i .

Since G_i is 2-connected, by induction every face boundary of G_i is a cycle. Since G_{i+1} is plane, the curve for P_i must be entirely inside one face f of G_i . Let the boundary of f be a cycle C .

The two endpoints of P_i divide C into two paths Q_1 and Q_2 . Then the face f is divided into two faces bounded by the cycles

$$Q_1 \cup P_i \quad \text{and} \quad Q_2 \cup P_i.$$

All other face boundaries remain cycles. Therefore every face boundary of G_{i+1} is a cycle.

Hence every face boundary of G is a cycle. $\square$

Corollary 10.7 (Corollary 49). In a 3-connected plane graph, all neighbours of a vertex v lie on a common cycle not using v .

Proof. Let G be a 3-connected plane graph and let $v \in V(G)$. Then $G - v$ is a 2-connected plane graph. Consider the face of $G - v$ that contains the point where v was before its removal. By Theorem 48, this face is bounded by a cycle.

All neighbours of v in G must lie on this face boundary. Therefore all neighbours of v lie on a common cycle of $G - v$, so the cycle does not use v . $\square$

Example 10.8. The conclusion of Corollary 49 is not necessarily true for a 2-connected plane graph.

10.3 Faces and cycle spaces

Proposition 10.9 (Proposition 50). The cycle space of a 2-connected plane graph is spanned by the set of all facial cycles.

Proof. Any element of $\mathcal{C}(G)$ is a sum of cycles, so it suffices to prove that every cycle is spanned by the facial cycles.

Let C be any cycle. Let S be the sum of all facial cycles in the interior of C . An edge in the interior of C has both sides inside C , so it is counted twice in S . Hence it is not in S over $\mathbb{F}_2$.

An edge of C has exactly one side inside C , so it is counted once in S . Therefore $S = C$. Hence C is the sum of facial cycles. $\square$

Observation

The set of all facial cycles contains a basis for $\mathcal{C}(G)$. Each edge of G is in two facial cycles, so each edge is counted at most twice in this basis.

Definition 10.10 (2-basis). A **2-basis** of a subspace of $\mathcal{E}(G)$ is a basis $\mathcal{B}$ such that each edge is used in at most two elements of $\mathcal{B}$.

Theorem 10.11 (MacLane's characterization of planar graphs). A graph G is planar if and only if $\mathcal{C}(G)$ has a 2-basis.

Proof idea. This will be proved later. $\square$

11 Bridges

Definition 11.1 (Bridge of a subgraph). Let H be a subgraph of a connected graph G . A **bridge** of H is either

1. a component F of $G - V(H)$ together with the edges in $\delta_G(V(F))$ and their endpoints; or
2. an edge in $E(G) - E(H)$ that joins two vertices of H , together with its two endpoints. This is called a **trivial bridge**.

Example 11.2. In the lecture example, the labelled subgraphs 1, 2, 3, 4 are bridges, and bridge 3 is trivial.

Definition 11.3 (Vertices of attachment and internal vertices). For a bridge B of H , the vertices that are in both B and H are the **vertices of attachment**, abbreviated VOA. The other vertices in B are called **internal vertices**.

Definition 11.4 (k -bridge). A bridge is a **k -bridge** if it has exactly k vertices of attachment.

Definition 11.5 (Equivalent bridges). Two bridges are **equivalent** if they have the exact same set of vertices of attachment.

Note

In this course, most of the time, H is a cycle.

Definition 11.6 (Segments). Let C be a cycle and let B be a k -bridge of C . The k vertices of attachment of B partition C into k **segments**. These are the paths on C between consecutive vertices of attachment, with no internal vertex being a vertex of attachment.

Definition 11.7 (Avoid). Two bridges of a cycle C **avoid** each other if the vertices of attachment of one bridge are entirely inside one segment of the other bridge.

Definition 11.8 (Overlap). Two bridges of a cycle C **overlap** if they do not avoid each other.

Definition 11.9 (Skew). Two bridges of a cycle C are **skew** if there exist distinct vertices v_1, v_2, v_3, v_4 on C in cyclic order such that v_1, v_3 are vertices of attachment of one bridge, and v_2, v_4 are vertices of attachment of the other bridge.

Example 11.10. In the lecture example, bridge 1 is a 3-bridge and has three segments. Bridges 1 and 2 avoid each other. Bridges 1 and 3 overlap, and in fact they are skew.

Proposition 11.11 (Proposition 51). Overlapping bridges of a cycle are either skew or equivalent 3-bridges.

Proof. Let B_1 and B_2 be overlapping bridges of a cycle C . Then both bridges have at least two vertices of attachment.

Suppose first that B_1 and B_2 are not equivalent. Then there exists a vertex of attachment u of B_1 that is not a vertex of attachment of B_2 . The vertex u lies in some segment of B_2 , say the segment from x to y .

Since B_1 and B_2 overlap, there exists a vertex of attachment v of B_1 in a different segment of B_2 . Then x, u, y, v occur in cyclic order on C , with u, v vertices of attachment of B_1 and x, y vertices of attachment of B_2 . Therefore B_1 and B_2 are skew.

Now suppose B_1 and B_2 are equivalent k -bridges. If $k = 1$ or $k = 2$, then they avoid each other. If $k = 3$, then they overlap and are equivalent 3-bridges. If $k \geq 4$, then they are skew.

Therefore overlapping bridges are either skew or equivalent 3-bridges. $\square$

12 Bridges in plane graphs

12.1 Inner and outer bridges

Given a connected plane graph G and a cycle C , each bridge of C is connected. By the Jordan Curve Theorem, each bridge is either entirely in $\text{Int}(C)$ or entirely in $\text{Ext}(C)$. Such a bridge is called an **inner bridge** or an **outer bridge**, respectively.

Theorem 12.1 (Theorem 52). If G is a connected plane graph and C is a cycle, then all inner bridges avoid each other. Similarly, all outer bridges avoid each other.

Preparation for the proof

1. Recall Corollary 46: any subdivision of K_5 or $K_{3,3}$ is not planar.

Definition 12.2 (Branch vertex). In a subdivision, the **branch vertices** are the vertices with degree at least 3.

Proposition 12.3 (Proposition 53). Let C be a simple closed curve. Let $p_1, \dots, p_k$ be k distinct points on C . Then for a point $x \in \text{int}(C)$, there exist k curves from x to $p_1, \dots, p_k$ such that the curves are in $\text{int}(C)$ except at the p_i 's, and they do not intersect except at x . The same statement holds for $\text{ext}(C)$.

Proposition 12.4 (Proposition 54). Let G be connected. Let H be obtained from G by adding three new vertices x_1, x_2, x_3 and joining each x_i to a vertex $v_i \in V(G)$, where the v_i 's are not necessarily distinct. Then there exists $v \in V(G)$ such that there is a 3-fan from v to $\{x_1, x_2, x_3\}$ in H .

Proof of Theorem 52. We prove the statement for inner bridges. The proof for outer bridges is similar.

First remove all outer bridges except for their vertices of attachment, so that C is the boundary of the outer face.

Suppose two inner bridges B_1 and B_2 overlap. By Proposition 51, they are either skew or equivalent 3-bridges.

First suppose B_1 and B_2 are skew. Then there exist vertices of attachment v_1, v_3 of B_1 and v_2, v_4 of B_2 such that v_1, v_2, v_3, v_4 occur in cyclic order on C .

Pick a point $x \in \text{ext}(C)$. By Proposition 53, we can find four curves from x to v_1, v_2, v_3, v_4 in $\text{ext}(C)$, intersecting only at x . Treat these curves as a new vertex x and a 4-fan from x to $\{v_1, v_2, v_3, v_4\}$.

Since B_1 is connected, there is a v_1, v_3 -path P_1 in B_1 . Since B_2 is connected, there is a v_2, v_4 -path P_2 in B_2 . Then the cycle C , the paths P_1, P_2 , and the 4-fan from x form a subdivision of K_5 with branch vertices

$$x, v_1, v_2, v_3, v_4.$$

This subdivision is drawn in the plane, contradicting Corollary 46.

Now suppose B_1 and B_2 are equivalent 3-bridges. Let v_1, v_2, v_3 be their common vertices of attachment. By Proposition 54, there exists an internal vertex y_1 in B_1 such that there is a 3-fan from y_1 to $\{v_1, v_2, v_3\}$. Similarly, there exists an internal vertex y_2 in B_2 such that there is a 3-fan from y_2 to $\{v_1, v_2, v_3\}$.

By Proposition 53, add a new 3-fan outside C from a new vertex x to $\{v_1, v_2, v_3\}$. These three 3-fans together form a subdivision of $K_{3,3}$ with bipartition

$$\{v_1, v_2, v_3\} \quad \text{and} \quad \{x, y_1, y_2\}.$$

This also contradicts Corollary 46.

Therefore no two inner bridges overlap, so all inner bridges avoid each other. The same argument applies to outer bridges. $\square$

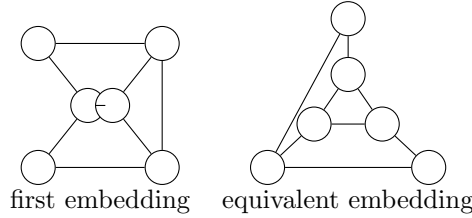
13 3-connected planar graphs

A planar graph may have several different planar embeddings. For general planar graphs, the facial boundaries can depend on the chosen embedding. For 3-connected planar graphs, however, the embedding is essentially unique.

Definition 13.1 (Equivalent embeddings). Two planar embeddings of a graph are **equivalent** if they have the same set of edges as face boundaries.

Definition 13.2 (Unique embedding). A planar graph has a **unique embedding** if every two planar embeddings of the graph are equivalent.

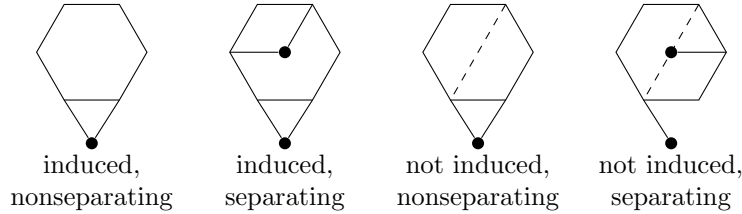
Example 13.3. The following two drawings represent equivalent embeddings: although the pictures look different, the same edge sets bound the faces.



Definition 13.4 (Induced nonseparating cycle). A cycle C of a graph G is an **induced nonseparating cycle** if

1. C is induced, meaning that $G[V(C)]$ has no edges other than the edges of C ; and
2. $G - V(C)$ is connected.

Example 13.5. The four possibilities are illustrated schematically below. A dashed chord indicates that the cycle is not induced, while vertices placed on both sides of the cycle indicate that deleting the cycle disconnects the graph.



Theorem 13.6 (Theorem 55). Let G be a 3-connected plane graph. A cycle C is a facial cycle if and only if C is an induced nonseparating cycle of G .

Proof. ($\Leftarrow$) Suppose that C is induced and nonseparating. Assume for a contradiction that C is not a facial cycle. Then C has both an inner bridge B_1 and an outer bridge B_2 .

Neither B_1 nor B_2 is trivial, because a trivial bridge would be a chord of C , contradicting the assumption that C is induced. Thus each of B_1 and B_2 has

an internal vertex. Since one bridge lies inside C and the other lies outside C , their internal vertices belong to different components of $G - V(C)$. Hence C is separating, a contradiction. Therefore C is facial.

($\Rightarrow$) Suppose that C is a facial cycle. By Proposition 47, we may redraw G so that C is the boundary of the outer face. Therefore all bridges of C are inner bridges, and by Theorem 52 they avoid one another.

First suppose that C is not induced. Then there is a trivial bridge xy , which is a chord of C . The vertices x and y divide C into two x, y -paths P_1 and P_2 . Every other bridge of C avoids the bridge xy , so all of its vertices of attachment lie entirely on P_1 or entirely on P_2 . Since G is 3-connected, each of P_1 and P_2 has an internal vertex. Consequently, removing x and y separates the internal vertices of P_1 from those of P_2 , together with the bridges attached to the two paths. Thus $\{x, y\}$ is a separating set, contradicting 3-connectivity. Hence C is induced.

Now suppose that C is separating. Then at least two nontrivial bridges B_1 and B_2 of C contain internal vertices in different components of $G - V(C)$. Since all bridges avoid one another, all vertices of attachment of B_2 lie on a single segment of B_1 , say the segment from x to y . Removing x and y separates an internal vertex of B_2 from an internal vertex of B_1 . Again $\{x, y\}$ is a separating set of size 2, a contradiction. Therefore C is nonseparating. $\square$

Corollary 13.7 (Corollary 56). Every 3-connected planar graph has a unique embedding.

Proof sketch. By Theorem 55, the facial cycles in any planar embedding are exactly the induced nonseparating cycles of G . Being induced and nonseparating are structural properties of the abstract graph, independent of the drawing. Thus every planar embedding has the same facial cycles, and therefore the same edge sets as face boundaries. Hence all embeddings are equivalent. $\square$

14 Preparing for Kuratowski's theorem

Definition 14.1 (Kuratowski subgraph). A subdivision of K_5 or $K_{3,3}$ is called a **Kuratowski subgraph**, abbreviated **KS**.

Theorem 14.2 (Kuratowski's theorem). A graph is planar if and only if it contains no Kuratowski subgraph.

The easy direction was proved earlier: a graph containing a subdivision of K_5 or $K_{3,3}$ is not planar. The hard direction is

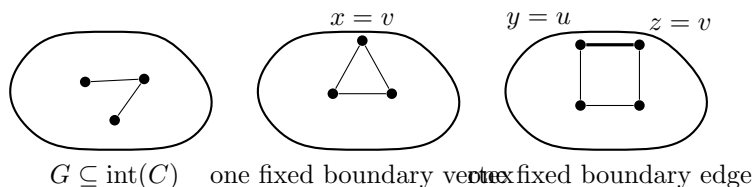
$$G \text{ is not planar} \implies G \text{ contains a Kuratowski subgraph.}$$

14.1 Flexibility in placing planar embeddings

Let G be a plane graph and let C be a simple closed curve.

1. It is possible to embed G entirely inside $\text{int}(C)$.

2. Let v be a point on the outer face of G , and let x be a point on C . Then G can be embedded in $\text{Int}(C)$ so that $v = x$ is the only intersection of the embedding with C .
3. Let uv be an edge on the outer face of G , and let yz be a curve contained in C . Then G can be embedded in $\text{Int}(C)$ so that the drawing of uv is exactly yz and is the only intersection of the rest of the embedding with C .



14.2 A fact about minimal separating sets

Proposition 14.3 (Proposition 57). If X is a minimal separating set of a graph G , then every vertex in X is adjacent to at least one vertex in every component of $G - X$.

Proof. Let $x \in X$ and let H be a component of $G - X$. Suppose that x has no neighbour in H . Since X is minimal, $X - \{x\}$ is not a separating set, so $G - (X - \{x\})$ is connected. However, in this graph the component H has no edge to x and no edge to any other component of $G - X$. Thus H remains separated from the rest of the graph, a contradiction. $\square$

15 Proof of Kuratowski's theorem

We prove the hard direction by induction on $|E(G)|$. Graphs with at most six edges can be checked directly to be planar. We may also assume that G has no isolated vertices, because deleting isolated vertices does not affect planarity or the existence of a Kuratowski subgraph.

Assume that G is nonplanar and that the result holds for every nonplanar graph with fewer edges than G . We divide the proof according to the connectivity of G .

15.1 Part 1: G is not connected

At least one component H of G is nonplanar; otherwise, planar embeddings of the components could be placed in disjoint regions of the plane. Since G has no isolated vertices and has more than one component, H has fewer edges than G . By induction, H contains a Kuratowski subgraph, which is also a Kuratowski subgraph of G .

Hence we may assume from now on that G is connected.

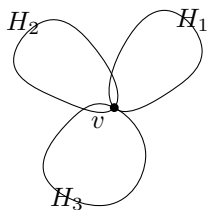
15.2 Part 2: G has a cut-vertex

Suppose that v is a cut-vertex of G , and let

$$H_1, \dots, H_k$$

be the bridges of the one-vertex subgraph $\{v\}$. Since $k \geq 2$, every H_i has fewer edges than G .

If some H_i is nonplanar, then by induction H_i contains a Kuratowski subgraph, and we are done. Otherwise, every H_i is planar. By Proposition 47, embed each H_i so that v lies on its outer face. Place v at a fixed point and divide a small neighbourhood of v into k pairwise disjoint regions. Embed each H_i in one region, with its copy of v at the common point. This gives a planar embedding of G , a contradiction.



15.3 Part 3: G has a separating set of size 2

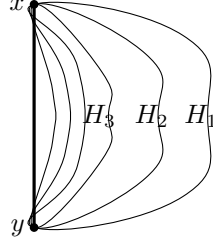
Assume that G is 2-connected and that

$$S = \{x, y\}$$

is a separating set. Since G is 2-connected, S is a minimal separating set. Let $H_1, \dots, H_k$ be the bridges of S , with the edge xy added to each bridge even when $xy \notin E(G)$.

Suppose first that some H_i is nonplanar. Since H_i has fewer edges than G , induction gives a Kuratowski subgraph K in H_i . If either $xy \in E(G)$ or K does not use xy , then K is already a Kuratowski subgraph of G . The only remaining possibility is that $xy \notin E(G)$ and the added edge xy is used by K . Choose another bridge H_j . Since H_j is connected and contains x and y , it contains an x, y -path P . Replacing the edge xy in K by P produces a subdivision of the same graph entirely inside G .

Now assume that every H_i is planar. Each H_i has a planar embedding in which xy is on the outer face. Start with an embedding of H_1 . Inductively, after embedding $H_1, \dots, H_i$, choose a face whose boundary contains xy and place H_{i+1} inside that face so that the two drawings share only the edge xy . At the end, delete the added edge xy if it was not an edge of G . This gives a planar embedding of G , a contradiction.



15.4 Part 4: G is 3-connected

By Theorem 29, there exists an edge

$$e = xy$$

such that G/e is 3-connected. Let z denote the vertex of G/e obtained by contracting x and y .

15.4.1 Part 4a: G/e is not planar

By induction, G/e contains a Kuratowski subgraph H . If $z \notin V(H)$, then H is also a subgraph of G , and we are done. Assume that $z \in V(H)$.

Case 1: z is not a branch vertex of H . Then $d_H(z) = 2$. Let u and v be the two neighbours of z in H . When the contraction is undone, each of u and v is adjacent in G to at least one of x and y .

If both u and v are adjacent to x , replace the path $u-z-v$ in H by $u-x-v$. The same argument applies if both are adjacent to y . Otherwise, after possibly interchanging u and v , we have

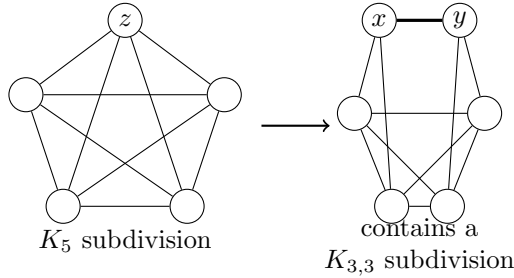
$$ux, yv \in E(G).$$

Replace $u-z-v$ by the path $u-x-y-v$. In every case, we obtain a Kuratowski subgraph of G .

Case 2: z is a branch vertex of H . Let $S = N_H(z)$. If every vertex of S is adjacent in G to x , replace z by x ; similarly, if every vertex of S is adjacent to y , replace z by y .

Suppose next that x is adjacent to every vertex of S except one vertex w . Then w must be adjacent to y . Use x as the branch vertex and replace the branch edge from z toward w by the path $x-y-w$. The symmetric case is identical.

The only remaining case is that x and y are each adjacent to exactly two distinct vertices of S . Then $|S| = 4$, so H is a subdivision of K_5 . Undoing the contraction splits the branch vertex z into adjacent vertices x and y , each joined to two of the remaining four branch vertices. The resulting graph contains a subdivision of $K_{3,3}$. Therefore G again contains a Kuratowski subgraph.



15.4.2 Part 4b: G/e is planar

Since G/e is 3-connected and planar, Corollary 49 implies that all neighbours of z lie on a common cycle C in $(G/e) - z$. In G , every neighbour of z is adjacent to at least one of x and y . Let H_x and H_y be the bridges of C in $G - xy$ that contain x and y , respectively.

Suppose first that H_x and H_y avoid each other. Embed C as a simple closed curve. Place x in $\text{int}(C)$ and join x to its neighbours on C by pairwise nonintersecting curves. The vertices of attachment of H_y all lie on one segment P determined by consecutive vertices of attachment u and v of H_x . The curve

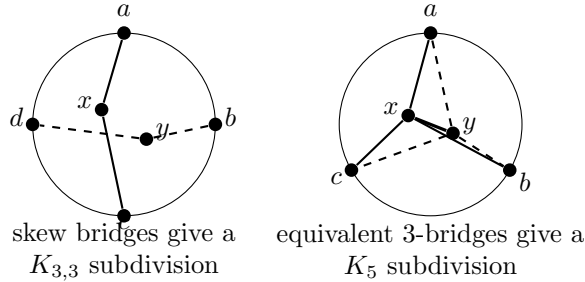
$$P \cup ux \cup xv$$

bounds a region. Place y in that region and join it without crossings to x and to all its neighbours on P . This gives a planar embedding of G , a contradiction.

Therefore H_x and H_y overlap. By Proposition 51, they are either skew or equivalent 3-bridges.

- If H_x and H_y are skew, choose alternating vertices of attachment on C . The corresponding paths through x and y , together with the appropriate segments of C , form a subdivision of $K_{3,3}$.
- If H_x and H_y are equivalent 3-bridges, let their common vertices of attachment be a, b, c . The three segments of C form a subdivided triangle on a, b, c ; both x and y are joined to each of a, b, c , and $xy \in E(G)$. These paths form a subdivision of K_5 with branch vertices x, y, a, b, c .

Both alternatives produce a Kuratowski subgraph, completing the final case and hence the proof of Kuratowski's theorem.



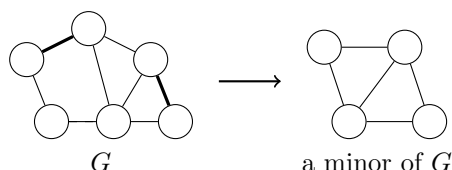
16 Wagner's theorem

Definition 16.1 (Minor). A graph H is a **minor** of a graph G if H can be obtained from G by a sequence of vertex deletions, edge deletions, and edge contractions.

Vertices in a minor

Each vertex of a minor H corresponds to a set of vertices of the original graph G that induces a connected subgraph of G . These connected sets are often called the **branch sets** of the minor model. Distinct vertices of H correspond to disjoint branch sets, and adjacent vertices of H have at least one edge of G joining their branch sets.

Example 16.2. In the schematic below, deleting the dashed edges and contracting each shaded connected set to one vertex produces a smaller graph as a minor.



Lemma 16.3 (Lemma 58). If G contains a subdivision of H , then H is a minor of G .

Proof. Let H' be a subdivision of H contained in G . Delete all vertices and edges outside H' . Each edge of H corresponds to a path in H' . Contract all but one edge of each such path. The resulting graph is isomorphic to H , so H is a minor of G . $\square$

Example 16.4. A subdivision of K_4 contains K_4 as a minor: contract the subdivided paths until each becomes a single edge.

Theorem 16.5 (Theorem 59: Wagner's theorem). A graph is planar if and only if it has neither K_5 nor $K_{3,3}$ as a minor.

Proof idea. If a graph is planar, then every graph obtained from it by deletions and contractions is planar. Since K_5 and $K_{3,3}$ are nonplanar, neither can be a minor of a planar graph.

Conversely, if G is nonplanar, Kuratowski's theorem gives a subdivision of K_5 or $K_{3,3}$ in G . By Lemma 58, the corresponding graph is a minor of G . $\square$

Kuratowski vs Wagner

Let K be all graphs without K_5 or $K_{3,3}$ subdivision.

Let W be all graphs without K_5 or $K_{3,3}$ minor.

Then $W \subset K$. Wagner's theorem implies $W = K$

Theorem 16.6. A graph G has a subdivision of K_5 or $K_{3,3}$ if and only if G has K_5 or $K_{3,3}$ as a minor.

Proof. ($\Leftarrow$) Suppose G has a $K_{3,3}$ minor. Using proposition, find 3-fans from each part. Now suppose G has a K_5 minor. If a 4-fan exists for each part, then we have K_5 subdivision. Find a 3-fan in one part. Join the 4th to the fan through a path. $\square$

17 Embedding on other surfaces

Theorem 17.1. (Graph minor theorem) Any infinite sequence of graphs $G_1, G_2, \dots$ include 2 graphs G_i, G_j where $i < j$ and G_i is a minor of G_j . (Proved by Seymour and Robertson, 20 papers)

Definition 17.2. A class of graphs $\mathbb{G}$ is minor-closed if any minor of any graph in $\mathbb{G}$ is in $\mathbb{G}$.

Example 17.3. Set of planar graphs. The set of all graphs embedded on a surface.

Definition 17.4 (minor-minimal). A graph G is minor-minimal for a minor-closed family $\mathbb{G}$ if $G \notin \mathbb{G}$ and any minor of G is in $\mathbb{G}$

Minor-minimal graphs for planar graphs

$K_5, K_{3,3}$

Theorem 17.5. For any minor-closed family of graphs $\mathbb{G}$, there is a finite number of minor-minimal graphs.

Proof. Suppose there are infinitely many of them, then we can put them in an infinite sequence, and one is a minor of the other by the Graph Minor Theorem. Contradiction. $\square$

Implication on embeddings on other surfaces

For any surface S , there exists finite set of graphs H where " G can be embedded on S if and only if G does not contain any graph in H as a minor".

Torus: H has size $\geq 17,500$

18 Algebraic characterization of planar graphs

Definition 18.1. A 2-basis of a subspace is a basis where each edge appears in at most 2 elements of the basis.

Theorem 18.2. A graph is planar if and only if its cycle space has a 2-basis.

Proof. ($\Rightarrow$) By last proposition, this is true for 2-connected graphs. $\mathcal{C}(G)$ has a 2-basis if and only if every block of G has a 2-basis. If a block is an

isolated vertex or a cut-edge, then it has a trivial cycle space, which has a trivial 2-basis. $\square$

Proposition 18.3. Suppose H is obtained from G by removing an isolated vertex, removing an edge, or contracting an edge. If $\mathcal{C}(G)$ has a 2-basis, then $\mathcal{C}(H)$ has a 2-basis.

Proof. Suppose G has c components. Let $\{E_1, \dots, E_k\}$ be a 2-basis for $\mathcal{C}(G)$, where $k = \dim \mathcal{C}(G) = |E(G)| - |V(G)| + c$,

1. Suppose v is an isolated vertex in G , and $H = G - v$. Then $\{E_1, \dots, E_k\}$ is a 2-basis of H .
2. Suppose $e \in E(G)$, and $H = G - e$. If e is a cut-edge, then e is not in any cycle, so e is not in any E_i . Then $\{E_1, \dots, E_k\}$ is a basis for $\mathcal{C}(H)$. Assume e is not a cut-edge. So e is in some E_i . If e is in exactly one element say E_k , then $\{E_1, \dots, E_{k-1}\}$ is a basis of $\mathcal{C}(H)$. Suppose e is in E_{k-1}, E_k , then $E_{k-1} + E_k$ does not contain e . So $\{E_1, \dots, E_{k-2}, E_{k-1} + E_k\}$ is a 2-basis of $\mathcal{C}(H)$.
3. Let $e \in E(G)$, $H = G/e$. Then $\{G - e, \dots, E_k - e\}$ (where $E_i - e = E_i$ if $e \notin E_i$) is a 2-basis of $\mathcal{C}(H)$

$\square$

19 Matchings

Definition 19.1 (matchings, covered, exposed). A **matching** M of a graph G is a subset of $E(G)$ that do not share any vertices. A vertex incident with an edge in M is **covered**, or **M-covered**. Otherwise it is **exposed**, or **M-exposed**.

Note

We can also think of M as a spanning subgraph of max deg 1.

Definition 19.2 (Maximum matching). A matching in G with the most number of edges. The size of a maximum matching is denoted $\alpha'(G)$

Definition 19.3 (Deficiency(for a graph)). Number of vertices exposed by a maximum matching. Denoted $def(G)$. [$def(G) = |V(G)| - 2\alpha'(G)$]

Definition 19.4 (Perfect matching / 1-factor). A matching that covers all vertices of G .

20 Matchings in bipartite graphs

20.1 Hall's Theorem

Hall's condition

[A necessary condition for the existence of a matching that covers A .]

For any $S \subset A$, such a matching matches each vertex in S to a distinct neighbour, so the neighbour set has size at least $|S|$.

Definition 20.1 (Neighbour set). For $S \subset V(G)$, its neighbour set is the set of all vertices in $V(G) - S$ adjacent to some vertex in S , denoted $N_G(S)$ or $N(S)$.

Theorem 20.2 (Hall's theorem). Let G be bipartite with bipartition (A, B) . Then G has a matching that covers all vertices in A if and only if for all $S \subset A$, $|N(S)| \geq |S|$.

Proof. ($\implies$) See above

($\impliedby$) We prove by induction on $|A|$. When $|A| = 1$, $|N(A)| = 1$. So the vertex in A is in a matching of size 1. Now assume $|A| \geq 2$. Two cases:

1. Suppose $|N(S)| > |S|$ for all nonempty proper subsets $S \subset A$. Let $xy \in E(G)$ where $x \in A, y \in B$. Let $G' = G - \{x, y\}$. Let $T \subset A - \{x\}, T \neq \emptyset$. By assumption, $|N_G(T)| > |T|$. But only y is removed from B in G' , so $|N_{G'}(T)|$ is either $|N_G(T)|$ or $|N_G(T)| - 1$. So $|N_{G'}(T)| \geq |T|$. By induction, there is a matching M' in G' that covers $A - \{x\}$. Then $M' \cup \{xy\}$ is a matching in G that covers A .
2. Suppose $|N(S)| = |S|$ for some nonempty proper subset S of A . We break G into 2 parts: Let $G_1 = G[S \cup N(S)]$, $G_2 = G - [S \cup N(S)]$. (Goal: Find matchings in G_1 and G_2) In G_1 , let $T \subset S$. By assumption, $|N_G(T)| \geq |T|$. But $N_G(T) \subset N(S)$ since $T \subset S$. So $|N_{G_1}(T)| \geq |T|$. By induction, there exists a matching M_1 in G_1 that covers S .

In G_2 , let $U \subset A - S$. We claim that $|N_{G_2}(U)| \geq |U|$. Suppose otherwise, and $|N_{G_2}(U)| < |U|$. Then

$$|N_G(S \cup U)| = |N_G(S)| + |N_{G_2}(U)| = |S| + |N_{G_2}(U)| < |S| + |U| = |S \cup U|$$

This contradicts our assumption for $S \cup U$ in G . By induction, there exists a matching M_2 in G_2 that covers $A - S$. Then $M_1 \cup M_2$ is matching in G that covers A .

□

20.1.1 Formula for $\alpha'(G)$ in a bipartite graph

What is the size of maximum matching in a bipartite graph?

A simple upper bound

$$\exists S, |S| > |N(S)|$$

Definition 20.3 (Deficiency (For a set of vertices)). For a set $S \subset V(G)$, the deficiency of S is

$$def(S) = |S| - |N(S)|$$

Corollary 20.4. For a bipartite graph with bipartition (A, B) ,

$$\alpha'(G) = |A| - \max\{def(S) : S \subset A\}$$

Completing the proof of Tutte's Theorem

(b) Claim: There is a matching between all of S and the odd components of $G - S$.

Obtain H by shrinking each odd component of $G - S$ into a vertex (contract all edges), remove edges in $G[S]$. Let T be the new vertices. Then H is bipartite with bipartition (S, T) , and $|S| = |T|$. [Here we want apply Hall's Theorem]. Let $X \subset T$, consider $N_H(X)$. If we remove $N_H(X)$ from G , then the odd components corresponding to X remain odd components, since $N_H(X)$ are the only vertices adjacent to them. By assumption, $|N_H(X)| \geq |X|$. By Hall's theorem, there is a 1-factor in H , which corresponds to a matching between S and its odd components in $G - S$, each vertex matched to a distinct odd component.

(c) Claim: Each odd component of $G - S$ has a matching that covers all but the one vertex matched to S .

Let C be an odd component of $G - S$, suppose $a \in V(C)$ is matched to S . Let $Z \subset V(C - a)$. [Goal: $o(C - a - Z) \leq |Z|$] Suppose by way of contradiction that $o(C - a - Z) > |Z|$. Since $|V(C) - a|$ is even, by Parity Lemma, $o(C - a - Z) \geq |Z| + 2$. Let $W = S \cup \{a\} \cup Z$. Then the odd components of $G - W$ are the odd components of $C - a - Z$, and odd components of $G - S$ except C . Then

$$o(G - W) = o(C - a - Z) + o(G - S) - 1 \geq |Z| + 2 + |S| - 1 = |Z| + |S| + 1 = |W|$$

By assumption, $o(G - W) \leq |W|$. So $o(G - W) = |W|$, contradicting the maximality of S . By induction, $C - a$ has a 1-factor. Applying to all odd components of $G - S$ plus the matching from above, we get a 1-factor of G .

21 Corollaries of Tutte's Theorem

21.1 Formula for $\alpha'(G)$

A simple upper bound

If $o(G - S) > |S|$, then a max matching must expose at least $o(G - S) - |S|$ vertices.

Definition 21.1 (Deficiency). For $S \subset V(G)$, the **deficiency** of S is $def(S) = o(G - S) - |S|$.

Corollary 21.2. For any G , $\alpha'(G) = \frac{1}{2}(|V(G)| - \max\{def(S) : S \subset V(G)\})$

21.2 Tutte sets

Definition 21.3 (Tutte set). A **Tutte set** is a set of vertices with maximum deficiency.

Relation between a Tutte set and maximum matching

Say a graph has def 2 (max matching exposes 2 vertices)

Definition 21.4 (Essential, avoidable). A vertex v is essential if every maximum matching covers v . (or, $\alpha'(G - v) = \alpha'(G) - 1$) It is avoidable if some maximum matching exposes v .

Lemma 21.5. Every vertex in a Tutte set is essential.

Proof. Let $S \subset V(G)$ be a Tutte set, let $x \in S$. Let $S' = S \setminus \{x\}$. Then $G - S = G' - S'$, and hence

$$o(G - S) = o(G' - S').$$

By the Tutte–Berge formula,

$$\begin{aligned} \text{def}(G') &\geq \text{def}_{G'}(S') \\ &= o(G' - S') - |S'| \\ &= o(G - S) - (|S| - 1) \\ &= o(G - S) - |S| + 1 \\ &= \text{def}_G(S) + 1 \\ &= \text{def}(G) + 1 \end{aligned}$$

Then $\alpha'(G') < \alpha'(G)$, so x is essential. $\square$

Theorem 21.6. Let S be a Tutte set in G . Then

1. Every vertex in S is essential, and is matched to a distinct odd component of $G - S$ in any maximum matching.
2. The even components of $G - S$ have 1-factor, and all vertices are essential.
3. Each odd component of $G - S$ has a near-perfect matching. (a matching that covers all but one vertex)

22 Corollaries of Tutte’s Theorem

22.1 A sufficient condition for a 1-factor

Corollary 22.1 (Petersen’s Theorem). A 3-regular graph with no cut-edge has a 1-factor.

Proof. Assume G is connected. Let $S \subset V(G)$, let $k = o(G - S)$. [Want: $|S| \geq k$] If $k = 0$, then $|S| \geq k$. Assume $k \geq 1$. Let $C_1, \dots, C_k$ be the odd components of $G - S$. Let F_i be the set of all edges joining C_i to S . The sum $\sum_{v \in V(C_i)} \deg(v) = 3|V(C_i)|$ since G is 3-regular. This sum is odd since C_i is an odd component. Now $\sum_{v \in V(C_i)} \deg(v) = 2|E(G[C_i])| + |F_i|$ Since the sum is odd and $2|E(G[C_i])|$ is even, $|F_i|$ is odd. Since G has no cut-edges, $|F_i| \geq 3$. Over all k components, $\sum_{i=1}^k |F_i| \geq 3k$. Each such edge has one end in S . Each vertex has degree 3, so $|S| \geq 3k/3 = k$. So $|S| \geq o(G - S)$. By Tutte’s theorem, there is a 1-factor. $\square$

23 Factor-critical graphs

Definition 23.1 (factor-critical). A graph G is factor-critical if $G - v$ has a 1-factor for all $v \in V(G)$. (Note a factor-critical graph is connected)

Theorem 23.2. Let G be a connected graph. Then G is factor-critical if and only if every vertex of G is avoidable.

Proof. ($\implies$) If G is factor-critical, then $G - v$ has a 1-factor, which is a maximum matching of G . So v is avoidable.

($\impliedby$) Suppose every vertex is avoidable. Then ϕ is the only Tutte set in G . Then $\text{def}(\phi) = o(G - \phi) - |\phi| = o(G) = 0$ or 1 since G is connected. Also $\text{def}(G) = \text{def}(\phi)$. If $\text{def}(G) = 0$, then G has a 1-factor, so all vertices are essential, contradiction. So $\text{def}(G) = 1$. By Tutte-Berge, $\alpha'(G) = \frac{1}{2}(|V(G)| - 1)$. Since v is avoidable, $\alpha'(G - v) = \alpha'(G) = \frac{1}{2}(|V(G)| - 1) = \frac{1}{2}(|V(G - v)|)$. So $G - v$ has a 1-factor, and G is factor-critical. $\square$

Theorem 23.3. If T is a maximal Tutte set, then each component of $G - T$ is factor-critical.

Proof. The proof is similar to part (c) of Tutte's Theorem. Removing a maximal Tutte set gives odd components by (a), there is a 1-factor when removing any vertex in the comp by (c). $\square$

24 Structure of factor-critical graphs

Definition 24.1 (odd closed-ear decomposition). An odd closed-ear decomposition of G is a sequence of graphs $G_0, G_1, \dots, G_k$ where

- G_0 is a vertex of G , $G_k = G$.
- For $0 \leq i \leq k - 1$, $G_{i+1} = G_i + P_i$ where P_i is either an ear of G_i of odd length, or an odd cycle that intersects G_i at exactly one vertex.

Theorem 24.2 (Lovasz). A graph G is factor-critical if and only if G has an odd closed-ear decomposition.

Proposition 24.3. Let M_1, M_2 be two matchings in G . Then the nontrivial components of $M_1 \triangle M_2$ are paths and even cycle, with edge alternating between M_1 and M_2 .

Now we prove the Lovasz's Theorem:

Proof. ($\impliedby$) Suppose G has an odd closed-ear decomposition, $G_0, \dots, G_k$. Induct on index of G_i .

Base case: G_0 is a vertex, factor-critical. Assume G_i is factor-critical. Let $G_{i+1} = G_i + P_i$. $\square$